

# SQL

## Theory Notes

### 1. SQL Overview

- **SQL (Structured Query Language)** → language for managing relational databases.
- Developed by **IBM in 1987**.
- Works with relational databases like MySQL, PostgreSQL, Oracle, SQL Server.

### 2. Types of SQL Commands

- **DDL (Data Definition Language)** → defines structure.
  - CREATE, ALTER, DROP, TRUNCATE.
- **DML (Data Manipulation Language)** → modifies data.
  - INSERT, UPDATE, DELETE.
- **DQL (Data Query Language)** → queries data.
  - SELECT.
- **DCL (Data Control Language)** → controls access.
  - GRANT, REVOKE.
- **TCL (Transaction Control Language)** → manages transactions.
  - COMMIT, ROLLBACK, SAVEPOINT.

### 3. DDL Examples

- Create table:

```
CREATE TABLE fiit(  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  name VARCHAR(100),  
  course VARCHAR(100),  
  email VARCHAR(100)  
);
```

- Alter table: add, modify, drop, rename columns.
- Drop table: deletes table structure.
- Truncate: clears all rows but keeps structure.

### 4. DML Examples

- Insert data:

```
INSERT INTO fiit VALUES(1, "java", "java@gmail.com", 6, 35000);
```

- Update data:

```
UPDATE fiit SET duration=10 WHERE id=2;
```

- Delete data:

```
DELETE FROM fiit WHERE course_name="java";
```

## 5. DQL Examples

- Select all:

```
SELECT * FROM fiit;
```

- Select with alias:

```
SELECT course_name AS Course, duration AS Duration FROM fiit;
```

- Filtering: WHERE, AND, OR.
- Sorting: ORDER BY.
- Aggregates: SUM, AVG, COUNT, MAX, MIN.
- Limit results: LIMIT.

## 6. Operators in SQL

- Arithmetic: +, -, \*, /, %.
- Comparison: =, !=, <, >, <=, >=.
- Logical: AND, OR, NOT.

## Practice Tasks

### Short Answer

1. What is the difference between DROP and TRUNCATE?
2. Why is ALTER used in SQL?
3. What does LIMIT do in a query?
4. How does SUM differ from COUNT?

### Coding Exercises

1. Create a table `students` with fields: `id`, `name`, `age`, `course`, `fees`.
2. Insert 5 records into `students`.
3. Update the fees of one student.
4. Delete a student record by ID.
5. Write a query to select students with fees greater than 30,000.
6. Find the average fees of all students.
7. Use ORDER BY to list students by fees descending.
8. Demonstrate LIMIT to fetch only top 2 students.
9. Write a query to count total students.
10. Drop the `students` table.

```

-- sql => structured query language
-- relational databases
-- IBM = > 1987
-- types of commands
-- DDL => data definition language
-- -- create, alter, drop , truncate

CREATE table fiit(id int AUTO_INCREMENT PRIMARY KEY,name VARCHAR(100),course
VARCHAR(100),email VARCHAR(100));

ALTER table fiit MODIFY name VARCHAR(50);

ALTER TABLE fiit ADD COLUMN duration INT;

ALTER TABLE fiit ADD COLUMN fees DOUBLE;

ALTER TABLE fiit DROP COLUMN course;

ALTER TABLE fiit CHANGE name course_name VARCHAR(50);

DROP TABLE fiit;

TRUNCATE TABLE fiit;

-- DML => data manipulation language
-- insert, update, delete
INSERT INTO fiit VALUES(1, "java", "java@gmail.com", 6 , 35000);
INSERT INTO fiit VALUES(2,"python", "python@gmail.com", 6 , 20000);

INSERT INTO fiit VALUES(3, "javascript", "javascript@gmail.com", 3, 15000);
INSERT INTO fiit VALUES(4,"mern_stack", "mern@gmail.com", 8, 45000);

UPDATE fiit SET duration= 10;

UPDATE fiit SET duration = 6 WHERE id=2;

DELETE FROM fiit WHERE course_name="java";

DELETE FROM fiit;

-- DQL => data query language

```

```

-- SELECT

SELECT * FROM fiit;

SELECT course_name AS Course ,duration AS Duration FROM fiit ;

SELECT * FROM fiit WHERE duration=6 LIMIT 1;

SELECT * FROM fiit WHERE fees>30000;

SELECT * FROM fiit ORDER BY fees DESC LIMIT 2;

SELECT * FROM fiit WHERE fees<30000 AND duration=3;

SELECT SUM(fees) as total FROM fiit;

SELECT AVG(fees) as total FROM fiit;

SELECT COUNT(*) AS COL_COUNT FROM fiit;

SELECT MAX(fees) as max_fees FROM fiit;

SELECT MIN(fees) as min_fees FROM fiit;

select 4+6 as Addresult

SELECT 10-5 as SubResult;

SELECT 10*5 as MulResult;

SELECT 10/5 as DivResult;

SELECT 10%5 as modulusResult;

-- DCL => Data control language
-- thought in next class
-- TCL => Transction control language
-- thought in next class

```